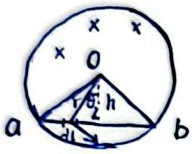
20.1

$$\varepsilon_{ab}=\int_a^b(\vec{v}\times\vec{B})\cdot d\vec{r}$$
$$=-\int_a^b Bv\,dr$$
$$=-\int_d^{d+l} Bv\,dr$$
$$=-v\int_d^{d+l}\frac{\mu_0 I}{2\pi r}\,dr$$
$$=-\frac{\mu_0 I v}{2\pi}\ln\frac{d+l}{d}$$
$$=-\frac{4\pi\times10^{-7}\times5\times10}{2\pi}\ln\frac{10+20}{10}$$
$$=-1.1\times10^{-5}\ V\quad<0$$

即动生电动势为 1.1×10^{-5} V 且a端电势高

20.5



在金属杆上取一段微元 dl

$$\varepsilon_{ab}=\int_a^b\frac{r}{2}\frac{dB}{dt}d\vec{l}$$
$$=\int_a^b\frac{r}{2}\frac{dB}{dt}\cdot\cos\theta\,dl$$
$$=\int_0^L\frac{h}{2}\frac{dB}{dt}\,dl$$
$$=\frac{h}{2}\cdot\frac{dB}{dt}\int_0^L dl$$
$$=\frac{\sqrt{R^2-\frac{L^2}{4}}\cdot L}{2}\frac{dB}{dt}\quad>0$$

即感生电动势为 $\frac{\sqrt{R^2-\frac{L^2}{4}}\cdot L}{2}\frac{dB}{dt}$，且b端电势高

20.11

$$\frac{mv^2}{R}=evB\ (\text{沿法向})\quad ①$$
$$m\frac{dv}{dt}=eE\ (\text{沿切向})\quad ②$$

由①可得 $v=\frac{eBR}{m}$ 代入②得

$$m\frac{d\frac{eBR}{m}}{dt}=eE\quad 即\quad \frac{dB}{dt}=\frac{E}{R}$$

~~又 $E=\frac{R\,d}{2}$~~

$$又\ E=\frac{1}{2\pi R}\frac{d\phi}{dt}=\frac{1}{2\pi R}\frac{\pi R^2 d\bar{B}}{dt}=\frac{R}{2}\frac{d\bar{B}}{dt}$$

则 $\frac{dB}{dt}=\frac{d\bar{B}}{2dt}$

即 $dB=\frac{d\bar{B}}{2}$

综上证毕

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20.15 (1) $L=\frac{N\phi}{I}=\frac{N\cdot\mu_0 nIS}{I}=N\cdot\mu_0 nS$

$=N\cdot\mu_0\cdot\frac{N}{l}S$

$=\frac{\mu_0 N^2 S}{l}$

$=\frac{\mu_0 N^2\pi R^2}{l}=\frac{4\pi\times10^{-7}\times1200^2\times\pi\times(2\times10^{-2})^2}{0.3}=7.6\times10^{-3}H$

(2) $\varepsilon=L\frac{dI}{dt}=7.6\times10^{-3}\times3\times10^2=2.3V$

20.17 (1) $L=\frac{N\phi}{I}$ $\boxed{\cancel{=\frac{N\mu_0 nI}{I}}}$

$d\phi=BdS=\mu_0 nIdS=\frac{\mu_0 NI}{2\pi r}hdr$

则 $\phi=\int d\phi=\int_{R_1}^{R_2}\frac{\mu_0 NI}{2\pi r}hdr$

$=\frac{\mu_0 NIh}{2\pi}\ln\frac{R_2}{R_1}$

则 $L=\frac{\mu_0 N^2 h}{2\pi}\ln\frac{R_2}{R_1}$

(2) $\boxed{\cancel{M_{21}=\frac{\phi}{I}=\frac{\mu_0 Nh}{2\pi}\ln\frac{R_2}{R_1}}}$

直导线的互感系数

$M_{12}=\frac{\phi}{I}=\frac{\mu_0 Nh}{2\pi}\ln\frac{R_2}{R_1}$

螺线管的互感系数

$M_{21}=\frac{N\phi'}{I}$

$\phi'=\int_{R_1}^{R_2}\frac{\mu_0 Ih}{2\pi r}dr=\frac{\mu_0 Ih}{2\pi}\ln\frac{R_2}{R_1}$

则 $M_{21}=\frac{\mu_0 Nh}{2\pi}\ln\frac{R_2}{R_1}$

可知 $M_{12}=M_{21}$

20.25

$W_m=\int_V\frac{1}{2}\vec{B}\cdot\vec{H}dV$

$=\int_V\frac{B^2}{2\mu_0}dV=\int_S\frac{B^2}{2\mu_0}\cdot1\cdot dS$

$=\int_0^{R_1}\frac{1}{2\mu_0}\left(\frac{\mu_0 Ir}{2\pi R_1^2}\right)^2 2\pi rdr+\int_{R_1}^{R_2}\frac{1}{2\mu_0}\left(\frac{\mu_0 I}{2\pi r}\right)^2 2\pi rdr$

$=\frac{1}{2\mu_0}\cdot\left(\frac{\mu_0 I}{2\pi}\right)^2 2\pi\left(\int_0^{R_1}\frac{r^3}{R_1^4}dr+\int_{R_1}^{R_2}\frac{1}{r}dr\right)$

$=\frac{\mu_0 I^2}{4\pi}\left(\frac{1}{4}+\ln\frac{R_2}{R_1}\right)$

又 $W_m=\frac{L}{2}I^2$

$L=\frac{2W_m}{I^2}=\frac{\mu_0}{2\pi}\left(\frac{1}{4}+\ln\frac{R_2}{R_1}\right)$

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